ARDUINO

Lec3

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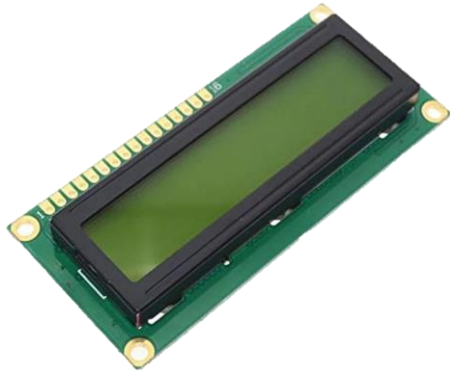
❑ Introduction

- In the Arduino applications , we can use a screen to display any data or a keypad to enter your data .
- In this lecture , we will learn about LCD and the keypad and how to connect them with Arduino .

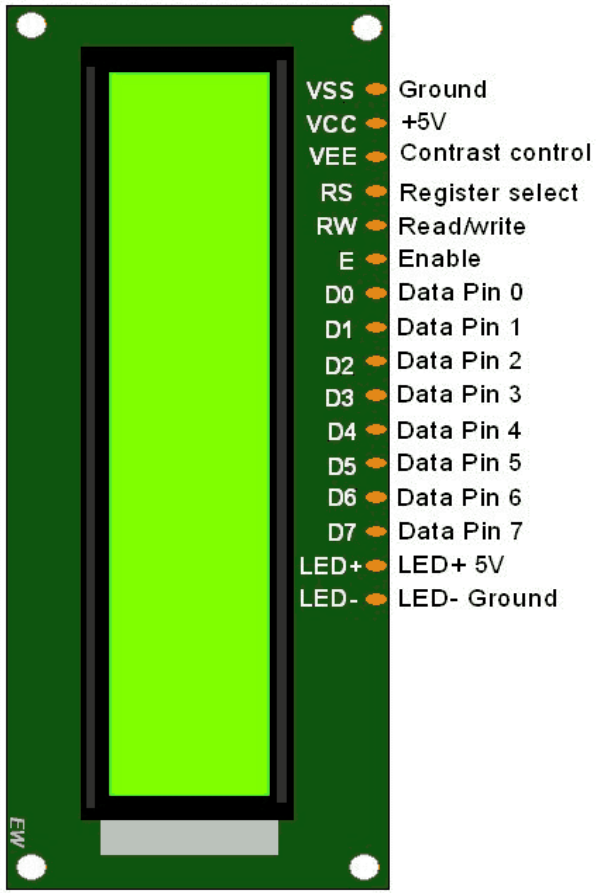


LCD

Keypad

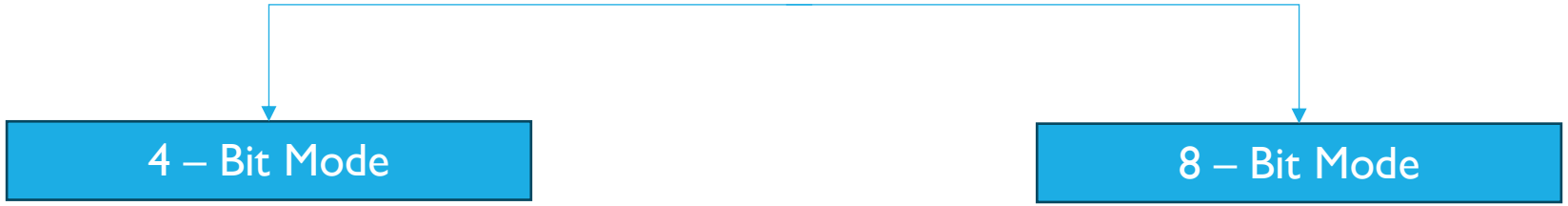
❑ LCD

- The LCD can be classified according to its size as :
  1. 2 * 16 LCD → 2 rows with 16 column .
  2. 4 * 16 LCD → 4 rows with 16 columns .
- As you see , the PCB consists of :
  i. 8 inputs (ascii - code from D0 to D7)
  ii. VSS is connected to the ground .
  iii. VDD or VCC is connected to VCC.
  iv. VEE (to change the lightness) is connected with a potentiometer .
  v. Rs to choose the mode of LCD (0→data , 1→ command) .
  vi. RW to choose if you want to read or write .
  vii. E is Enable .



❑ Modes of LCD

- There are 2 types of methods to load data :



Loading the first 4 bits from the data then the next 4 bits .

Loading the first 8 bits from the data in parallel .

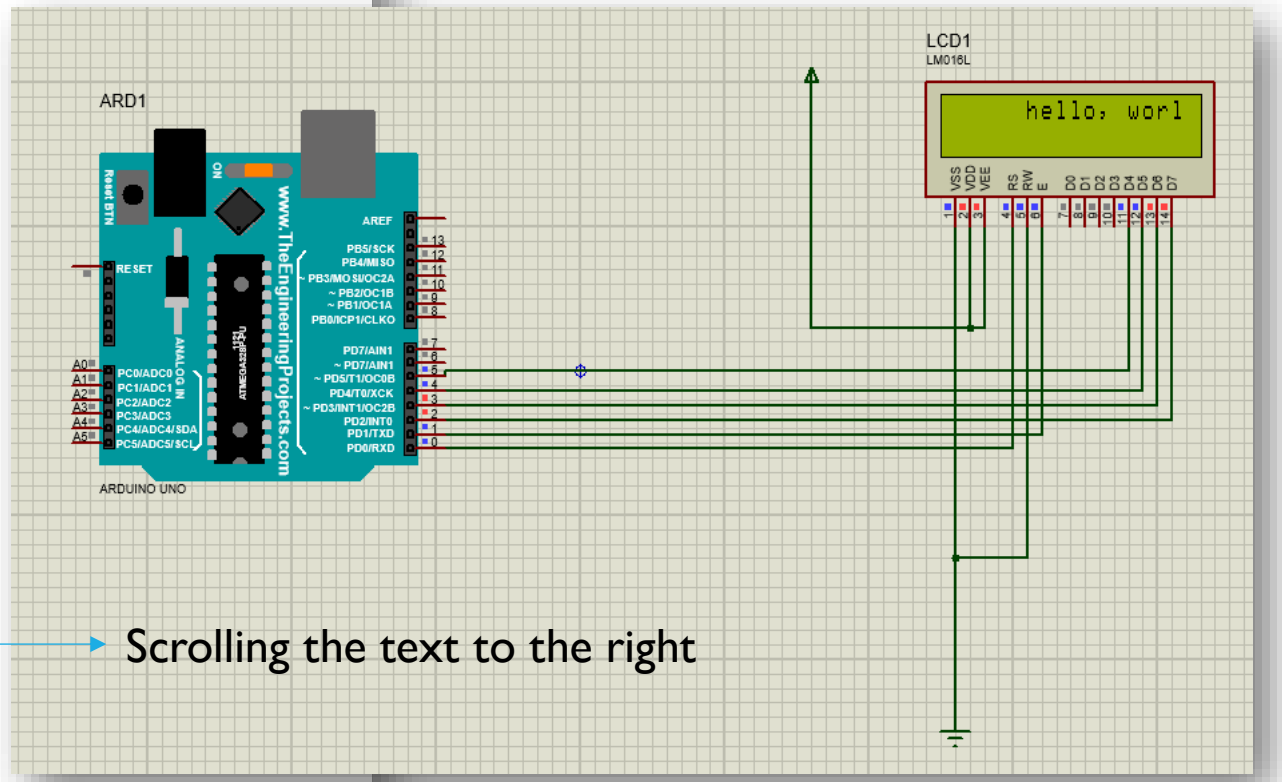
✓ Important Codes For LCD

- LiquidCrystal lcd (rs, en, d4, d5, d6, d7); → to define the pins of LCD which are connected to Arduino.
- lcd . begin (no. of cols , no. of rows) → to define the size of the LCD .
- lcd . print(req string) → to print any string .
- lcd . println (req string) → to print any string and moving to the next line .
- lcd . Blink ( ) → to blink the cursor .
- lcd . home( ) → to return the cursor to the starting position .
- lcd . clear ( ) → clear the data from the screen .
- lcd .setcursor (column , row) → to set the cursor .

Don't forget to import the module of LiquidCrystal .

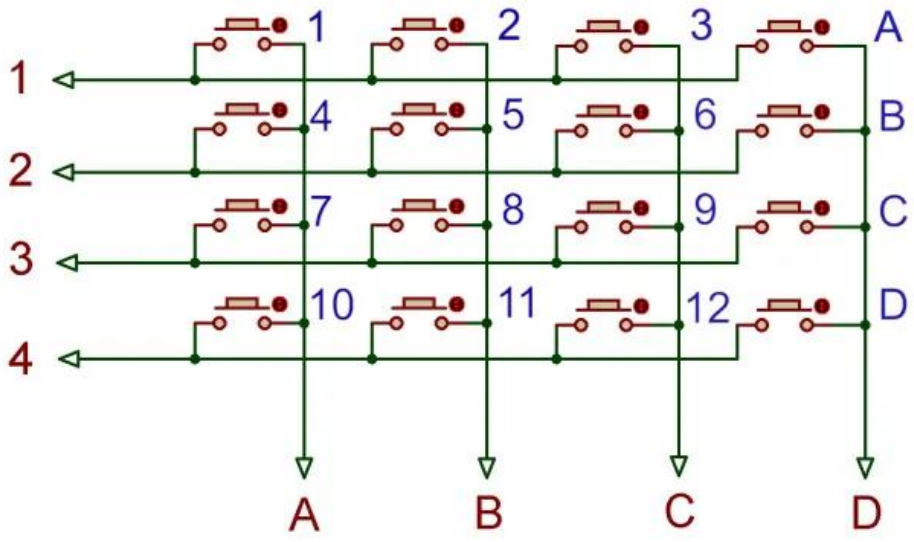
✓ Ex :- display a repeated “Hello , World” ?

```
// include the library code:
#include <LiquidCrystal.h>
// initialize the library by associating any needed LCD interface pin
// with the arduino pin number it is connected to
const int rs = 0, en = 1, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
int t1;
bool flag=true;
void setup() {
  // set up the LCD's number of columns and rows:
  lcd.begin(16, 2);
  // Print a message to the LCD.
  t1=millis();
}
void loop() {

  lcd.setCursor(0,0);
  if (flag){
    lcd.print("hello, world!");
    flag=false;
  }
  if ((millis()-t1)<1300){

    lcd.scrollDisplayRight();
    delay(100);
  }
  else{

    t1=millis();
    // lcd.home();
  }
}
```

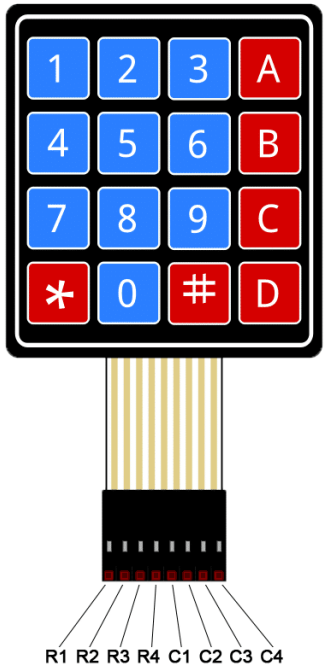


Scrolling the text to the right

❑ Keypad



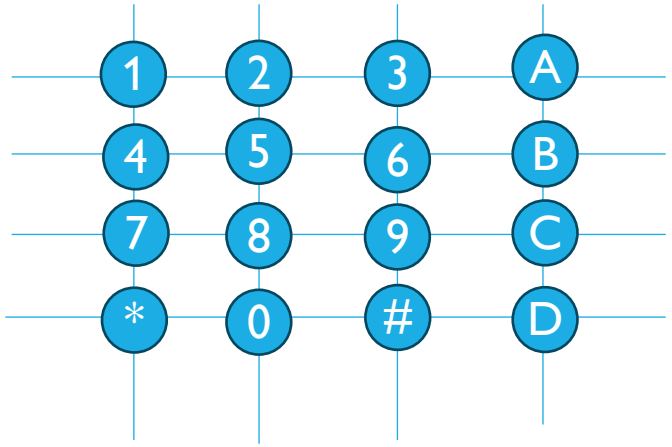
Internal structure



Keypad

Idea of the internal structure ??

- The keypad is a net of wires on each other such as in the right :
  - when pressing any button , 2 wires have been touched so that they will have the same potential at their windings .
  - The check method follows : set 0 on one row and 1 on the others when you press any button , the output potential for the column which the pressed button is in will be 0 .



SOME CODES

- At the first , import the module of keypad.h and to set a keypad in the following line .
- Keypad keypad = keypad(makeKeymap(keys) , pins_rows , pins_columns , no. of rows , no. of columns)
- To get the key which is pressed → keypad.getKey()
- In the 2nd line :
    1. we define no. of columns initially and for rows , too .
    2. we define pins_rows (Is an array of the pins of rows in Arduino .)
    3. We define pins_columns (Is an array of the pins of columns in Arduino .)
    4. Finally , we define keys'array .

→ The application for this codes in the following slide .

✓ Ex :- a program adding the input from keypad to lcd ??

```
#include <Keypad.h>
#include <LiquidCrystal.h>
const int ROW_NUM = 4; //four rows
const int COLUMN_NUM = 4; //three columns
char keys[ROW_NUM][COLUMN_NUM] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'}
};
byte pin_rows[ROW_NUM] = {A0,A1, A2, A3}; //connect to the row pinouts of the keypad
byte pin_column[COLUMN_NUM] = {12,11, 10, 9}; //connect to the column pinouts of the keypad
Keypad keypad = Keypad( makeKeymap(keys), pin_rows, pin_column, ROW_NUM, COLUMN_NUM );
const int rs = 0, en = 1, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);

void setup(){
  // Serial.begin(9600);
  lcd.begin(16, 2);
  // Print a message to the LCD.
}
void loop(){
  char key = keypad.getKey();

  if (key){
    lcd.print(key);
  }
}
```

THANK YOU

FOR PAYING ATTENTION